

Towards predictive engineering-type simulations of upward flame spread in SBI scenarios

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Abstract

Large eddy simulations of upward flame spread using FireFOAM are presented. Aiming at advancing predictive fire modelling, the approach considers the use of dynamic models, with limited use of model constants, for turbulence, combustion, and radiation. Modelling of convective heat transfer is based on Newton's law of cooling considering simplified correlations for natural convection. The thermal decomposition of the solid material is represented through a 1D pyrolysis model with optimised model-effective material properties. For validation purposes, medium-scale Single Burning Item (SBI) experiments are used, involving both inert materials (calcium silicate) and flammable walls involving both charring (MDF and plywood) and non-charring (PMMA) materials. Separate validations for the gas and solid phase are also presented. A detailed comparison between the CFD predictions and experimental data is performed, focusing on global parameters (i.e., HRR, mass loss rate, heat feedback) and local quantities (i.e., total heat fluxes). The modelling approach performs very well, with predictions being fairly grid-insensitive, showing relative differences in the predicted HRR of up to 47% between the simulations and the experiments. Convection contributes up to 30% of the total wall heat feedback, highlighting the importance of accurately modelling convection alongside radiation in early flame spread.

Keywords: flame spread, heat feedback, convection, radiation, FireFOAM, CFD

1. Introduction

To ensure fire safety, many building components must undergo testing and certification in accordance with established standards. For this purpose, fire tests are conducted to verify that a product complies with the minimum requirements specified in national building codes and regulations. These tests can vary in scale rang-

1 ing from bench scale tests up to large-scale façade tests. Fire testing can be subdi-
2 vided into reaction-to-fire testing, involving the determination of the flammability
3 and fire toxicity of materials, and fire resistance testing, designed to determine the
4 structural integrity of building components. The former, being the central focus
5 of the current paper, is a key aspect when it comes to fire growth and flame spread
6 over combustible surfaces. Evaluating the flammability behavior of materials re-
7 quires excessive fire testing which can often be time consuming and expensive.

8 Today, the fire behavior of building materials can also be analyzed using Com-
9 putational Fluid Dynamics (CFD). This method complements experimental test-
10 ing by providing additional insights and helping to reduce the number of physical
11 tests, thereby lowering the high costs associated with experimental campaigns.
12 Obviously this requires good accuracy and reliability of the CFD simulations. It
13 has been recently well recognised in the fire community that there is need for bet-
14 ter modelling of the physics in the near-wall region of boundary layer combustion
15 [2]. Such efforts are ongoing through the joint effort of the MaCFP (Measure-
16 ments and Computations of Fire Phenomena) workshop where both gas and con-
17 densed phase phenomena are considered and flame spread modelling is of great
18 interest. The present work aims at advancing (predictive) fire modelling when it
19 comes to engineering-type of flame spread simulations and to investigate the im-
20 portance of accurately predicting (convective and radiative) heat transfer in such
21 scenarios. A methodology of model validation (see Figure 1(a)) is needed in order
22 to develop a reliable and accurate CFD framework for modelling flame spread sce-
23 narios. Decoupling and validating the gas and solid phases separately is of great
24 importance in order to have confidence in the employed modelling approaches
25 and to avoid compensating errors in the modelling when complex scenarios are
26 considered. For this purpose, validation of the modelling approach against well-
27 characterized experiments in both phases is considered.

28 There is typically a wide range of length and time scales in fire scenarios in-
29 volving flame spread that need to be considered when employing CFD modelling.
30 Particularly in the context of both fire safety science and engineering, the consid-
31 ered sub-models need to be able to perform satisfactorily over a wide range of
32 grid sizes. A pre-requisite for obtaining accurate flame spread predictions over
33 different grids is having fairly grid-independent predictions in the gas phase. For
34 this purpose, the use of dynamic modelling approaches is appealing and has been
35 shown to perform satisfactorily over a wide range of grid sizes when it comes to
36 pool fire scenarios in the past [1]. Their application to flame spread scenarios is
37 therefore appealing and the main motivation for using them in the current CFD
38 study. It should be noted, however, that the application of dynamic models in sim-

1 ulations of flame spread scenarios is only expected to be meaningful as long as
2 the gas phase can be accurately resolved by the considered grid size. In addition,
3 an (often significant) increase in computational cost is expected from the use of
4 such dynamic models, compared to traditional models using constant parameters.
5 Finally, there can potentially be increased chances of a diverging solution unless
6 proper bounds of the dynamically-determined model parameters are defined.

7 The experimental test cases exhibit increasing complexity and include pool
8 fires, inert wall fires, pyrolysis tests and medium-scale flame spread scenarios of
9 charring and non-charring materials. A fundamental understanding of the various
10 physical phenomena involved in flame spread scenarios is essential, as numerous
11 challenges must be overcome for developing accurate models for predictive fire
12 behavior. More specifically, there is an intricate coupling of the different physics
13 involved in the gas phase, which cannot be easily decoupled, and which provides
14 the necessary heat feedback to the fuel surface which determines the rate of solid
15 pyrolysis and the subsequent flame spread (see Figure 1(b)). Over- or under-
16 predicting the heat feedback to the surface of flammable walls can then lead to
17 faster or slower flame spread, respectively. Given the influence not only of the
18 modelling itself, but also of the grid resolution, on the predicted flame tempera-
19 tures and resulting heat fluxes, it becomes clear that accurate predictions of flame
20 spread with the use of CFD and pyrolysis models, over a wide range of grid sizes,
21 becomes extremely challenging. Regardless of the stair-step validation method-
22 ology considered in the present work, it has to be acknowledged that uncertain-
23 ties in the employed approach still exist (i.e., physics and phenomena that can be
24 (significantly) different between scenarios of different scales that will introduce
25 uncertainties in the modelling). As such model validation in a given (gas/solid
26 phase) scenario does not necessarily imply the same model performance/accuracy
27 in another similar scenario which includes comparable physics.

28 Building on the knowledge previously gained in similar scenarios, e.g., [3, 4],
29 the paper reports on LES simulations of fire growth scenarios and aims to en-
30 hance predictive fire modelling when it comes to simulating upward flame spread.
31 Within this context, several Single Burning Item (SBI) scenarios [5, 6, 7], involv-
32 ing both charring (MDF, plywood) and non-charring (PMMA) materials, are con-
33 sidered for validation purposes of the CFD simulations. The CFD code used in this
34 study is FireFOAM, developed by FM Global. A fully dynamic approach is used
35 for modelling turbulence, combustion and radiation. Convective heat transfer at
36 the burning surface is modelled based on Newton's law considering simplified cor-
37 relations for natural convection, while a single-step Arrhenius pyrolysis reaction,
38 combined with optimised model-effective material properties (derived via inverse

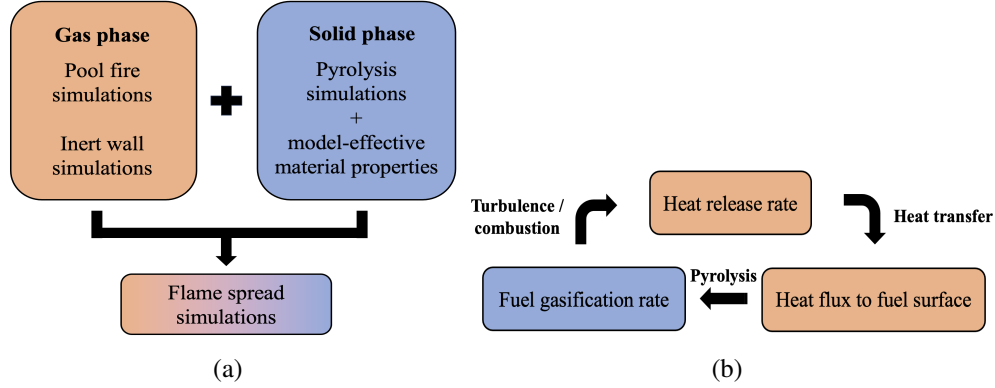


Figure 1: Illustration of the (a) methodology used for model validation and (b) gas-solid phase coupling in flame spread scenarios.

modelling and optimization approaches using FireFOAM [8]), is used for solid decomposition. As part of a stair-step validation exercise, scenarios involving just pool fires, inert walls (calcium silicate), and solid pyrolysis are also examined in order to validate the models in terms of predicted flame temperatures, total heat fluxes (i.e., heat feedback) on the inert walls and solid decomposition rates. A boundary layer analysis, for either momentum-driven or natural convection flows (i.e., a priori estimates of the size of the viscous sub-layer δ_{vsl}), suggests that grid resolutions in the order of mm are needed in order to have accurate predictions within the momentum ($\delta_{vsl} \approx \frac{\nu_w}{(\tau_w/\rho_w)^{0.5}}$) and thermal ($\delta_{vsl} \approx \frac{\alpha_w^{3/4}}{(\dot{q}_c''/(\rho_w c_{p,w}))^{1/4}(g\beta)^{1/4}}$) boundary layer [9]. The focus in the present study is explicitly on the use of grid sizes that are sufficiently small for resolving reasonably well the fire plume dynamics, but that are not fine enough to accurately resolve the boundary layer near the walls. The study thus targets more engineering-type of flame spread simulations, aims to identify potential deficiencies used in the modelling approach and to report on improvements that need to be made in order to enhance the predictive capabilities of CFD codes towards accurately simulating such scenarios.

2. Modelling

The simulations are performed with FireFOAM (<https://github.com/fireFoam-dev>), developed by FM Global. An overview of the governing equations considered in the code can be found in [10] while the various sub-models considered in the simulations are presented below.

2.1. Turbulence modelling

Turbulence is modelled with the dynamic Smagorinsky model [11]:

$$\mu_{sgs} = \max(\bar{\rho}(c_s\Delta)^2|\tilde{S}|, -\mu) \quad (1)$$

where Δ is the filter width (i.e., cube root of the cell volume), $\tilde{S}$ is the resolved strain rate and μ is the molecular dynamic viscosity. The dynamic model is chosen as to account for the weakly turbulent conditions often present at the base of wall fires and for the transition of the flow occurring further downstream [9]. In addition, back-scattering (i.e., reverse energy cascade) [12, 13], which occurs near the base of buoyant fires sources, is incorporated by allowing the sgs viscosity to take negative values (i.e., up to minimum $-\mu$) which results in $\mu_{eff} = \mu + \mu_{sgs} \geq 0$.

The sgs kinetic energy and sgs dissipation rate are estimated as [14]:

$$k_{sgs} = c_I \Delta^2 |\tilde{S}|^2 \quad (2)$$

$$\epsilon_{sgs} = \frac{c_\epsilon k_{sgs}^{3/2}}{\Delta} \quad (3)$$

The model parameters, c_s , c_I , and c_ϵ , as well as the turbulent Prandtl number, Pr_t , are calculated based on a dynamic procedure as local averages of their face values. No upper bounds are defined for the model parameters and a simple top-hat filter is used as a test filter while the ratio of the test filter to the LES filter is set to $\sqrt{6}$ [15]. More details regarding the dynamic procedure employed can be found in [10].

2.2. Combustion modelling

The combustion model considers a one-step, infinitely fast, chemical reaction, combined with the eddy dissipation model (EDM) [16] for modelling turbulence-chemistry interactions, and calculates the fuel reaction rate as:

$$\overline{\dot{\omega}}_F''' = -\bar{\rho} \frac{\min(\tilde{Y}_F, \tilde{Y}_{O_2}/s)}{\tau_{mix}} \quad (4)$$

where $\tilde{Y}_F$ and $\tilde{Y}_{O_2}$ are the fuel and oxygen mass fractions, respectively and s is the stoichiometric oxygen-to-fuel ratio.

Typical fire scenarios involve a wide range of time scales, e.g., associated with diffusive, advective and/or buoyant transport. Determination of the appropriate sgs mixing time scale will inherently depend on the levels of turbulence present

1 in the scenario (which in turn will affect the sgs dissipation rate on a given grid
 2 size). Therefore, the use of a dynamic approach, both in terms of estimating the
 3 sgs quantities through turbulence modelling and in determining the mixing time
 4 scale, is essential for accurate sgs mixing over a wide range of grid sizes. In this
 5 case, mixing under laminar (τ_{lam}) and turbulent (τ_{turb}) conditions is considered
 6 to be governed by diffusion and sgs advection, respectively. Two of the most
 7 important turbulent time scales are considered: the sgs velocity stretching time
 8 scale (Δ/u'), associated with large-scale mixing; and the Kolmogorov time scale
 9 $((\nu/\epsilon_{sgs})^{1/2})$, associated with the mixing on molecular level [17]. A synthetic
 10 turbulent mixing time scale is then based on their geometric mean. The mixing
 11 time scale, τ_{mix} , is then calculated as [17, 18]:

$$\tau_{mix} = \min\left(\underbrace{\frac{\Delta^2}{\alpha}}_{\tau_{lam}}, \underbrace{\sqrt{\frac{\Delta}{u'}\left(\frac{\nu}{\epsilon_{sgs}}\right)^{1/2}}}_{\tau_{turb}}\right) \quad (5)$$

12 where α is the thermal diffusivity and $u' = (2k_{sgs}/3)^{1/2}$ is the rms of the sgs
 13 turbulent velocity fluctuations.

14 A simple approach is used to account for soot modelling. The amount of soot
 15 produced is determined by the single step reaction of fuel and oxidizer through a
 16 prescribed soot yield (i.e., $y_s = 0.024$ [19] for C_3H_8) with its spatial distribution
 17 mapped on one of the combustion products (i.e., CO_2). The considered value is
 18 within the range of values reported for wood / PMMA in the literature [19]. This
 19 approach does not aim to quantitatively predict the soot concentrations but rather
 20 to simply account for the bulk effect of soot in radiation calculations, which can
 21 be important when sooty fuels are involved.

22 2.3. Radiation modelling

23 The radiative transfer equation (RTE) is solved by the finite volume discrete
 24 ordinates method with absorption modelled with the weighted-sum-of-gray-gases
 25 (WSGG) model [20]. The choice of the model is based on a compromise between
 26 accuracy (when compared to the gray gas model) and computational speed (when
 27 compared to a wide band model). Within WSGG, the total emissivity of a H_2O
 28 and CO_2 mixture is calculated as the sum of four fictitious gray gases ($N_g = 4$)
 29 weighted with a temperature dependent weighing factor as:

$$\epsilon = \sum_{i=0}^{N_g} \alpha_i (1 - e^{-\kappa_i p L}) \quad (6)$$

1 where N_g is the number of gray gases, α_i and κ_i (taken from [21]) are the emissiv-
 2 ity weighting factor and the absorption coefficient of the i th gray gas, respectively,
 3 p is the sum of the partial pressures of all absorbing gases, and L is the path length.
 4 The total absorption coefficient, including the contribution from soot, is then
 5 calculated as:

$$\kappa = \underbrace{-\frac{\ln(1 - \epsilon)}{L}}_{\text{gas}} + \underbrace{C_{soot} f_v T}_{\text{soot}} \quad (7)$$

6 where $f_v = (\bar{\rho} \tilde{Y}_s) / \rho_s$, $C_{soot} = 1862 \text{ m}^{-1} \text{K}^{-1}$ [22] and $\rho_s = 1800 \text{ kg/m}^3$ [23].

7 The path length is calculated as:

$$L = 3.6 \frac{V}{A} \quad (8)$$

8 where V and A are the volume and surface area, respectively, of the participating
 9 medium. The volume is calculated by summing all the cell volumes where re-
 10 action takes place, and the corresponding surface area is obtained by assuming a
 11 certain flame shape (depending on the burner geometry). The considered approach
 12 is capable of capturing the increase in the volume of the participating medium dur-
 13 ing the flame growth period which will also affect the absorption characteristics of
 14 the fire. This approach can be considered acceptable here given that no significant
 15 flame spread occurs (i.e., the fire is mainly concentrated near the corner). In other
 16 scenarios where significant flame spread can occur, the estimation of the surface
 17 area will ideally need to be made dynamically, resulting also in an increase the
 18 complexity of the calculation. For the calculation of the soot absorption coeffi-
 19 cient, only the soot produced by the burner is considered, not the soot produced
 20 from the solid fuel.

21 In general, the grid resolutions used for fire simulations are not capable of
 22 accurately resolving the sub-grid temperature fluctuations. Therefore, account-
 23 ing for turbulence-radiation interactions (TRI) is expected to be important when
 24 modelling the emission term. Similar to [24], the emission term can be approxi-
 25 mated as $\widetilde{\kappa T^4} \approx \tilde{\kappa} \tilde{T}^4 R_{T^4}$, assuming temperature self-correlation and a correlation
 26 between the filtered absorption coefficient and temperature. The correlation R_{T^4}
 27 can be approximated by expanding it and neglecting terms higher than second or-
 28 der as $R_{T^4} \approx 1 + 6C_{TRI}(\widetilde{T''^2} / \tilde{T}^2)$ with C_{TRI} accounting for the contribution of
 29 the neglected higher-order terms [25].

1 The divergence of the radiative heat fluxes is then estimated as follows:

$$\nabla \cdot \vec{q}_r'' = \underbrace{4\sigma\kappa\tilde{T}^4 \left(1 + 6C_{TRI} \frac{\widetilde{T''^2}}{\tilde{T}^2}\right)}_{\text{emission (TRI)}} + \underbrace{E\dot{\bar{Q}}_c'''}_{\text{correction}} - \underbrace{\kappa G}_{\text{absorption}} \quad (9)$$

2 where σ is the Stefan-Boltzmann constant, $\dot{\bar{Q}}_c'''$ is the heat release rate per unit
 3 volume, G is the total irradiance, the temperature variance is modelled as $\widetilde{T''^2} =$
 4 $c_z \Delta^2 \nabla |\tilde{T}|^2$ with c_z computed based on a dynamic procedure similar to [27], and
 5 E is a correction in the emission term:

$$E = \max\left(\chi_{r,exp} - \underbrace{\frac{\iiint_V (\text{emission} - \text{absorption}) dV}{\iiint_V \dot{\bar{Q}}_c''' dV}}_{\chi_{r,simulated}}, 0\right) \quad (10)$$

6 which ensures the correct amount of heat being released due to radiation when
 7 the flames are under-resolved (i.e., coarse-grid simulations). The consideration
 8 of $\max(\chi_{r,exp} - \chi_{r,simulated}, 0)$ ensures that no correction is applied if $\chi_{r,simulated}$
 9 exceeds $\chi_{r,exp}$ (i.e., the grid resolution is fine enough to accurately simulate the
 10 flame emission without an explicit correction).

11 In this work, $C_{TRI} = 2$ [26], which is a value that has been reported to be a
 12 good-fitting curve for ethanol and methanol pool fire data and for a quite different
 13 type of reacting flow as well (i.e., hydrogen-enriched oxy-methane flames). In the
 14 absence of available data for other scenarios and potentially other fuels, the value
 15 would be kept the same. Further model sensitivity on this model constant would
 16 be useful in the future.

17 The experimentally-determined (global) radiative fraction in the SBI scenarios
 18 involving flammable walls, only used as a correction factor in Eq. (10), is deter-
 19 mined based on the radiative fraction of the C_3H_8 burner, $\chi_{r,b}$, and that of the wall,
 20 $\chi_{r,w}$ (see Table 5), as:

$$\chi_{r,exp} = \frac{\dot{m}_b \chi_{r,b} + \dot{m}_w \chi_{r,w}}{\dot{m}_b + \dot{m}_w} \quad (11)$$

21 where $\dot{m}_b$ and $\dot{m}_w$ are the mass loss rates from the burner and the wall, respec-
 22 tively. The considered approach aims to capture any changes in the (global) ra-
 23 diative fraction, which will range between $\chi_{r,b}$ in the initial stage to $\chi_{r,w}$ when the
 24 fire grows in size, and apply the corresponding correction in the emission term.
 25 Note that $\chi_{r,exp} = \chi_{r,b}$ for the C_3H_8 pool fire scenarios or cases involving inert
 26 walls.

2.4. Convection modelling

Convective heat transfer is modelled based on Newton's law of cooling [28] expressed as:

$$\dot{q}_c'' = h \underbrace{(T - T_w)}_{\Delta T} \quad (12)$$

where T is the gas temperature in the first cell next to the wall and T_w is the wall temperature.

The use of typical empirical correlations related to convective heat transfer introduces a number of uncertainties when applied to fire scenarios [29] (e.g., not clearly defined free stream properties; different levels of turbulence compared to the conditions that the correlations were derived for; disturbance of the boundary level due to mass transpiration effects, etc). In addition, accurate determination of the thermophysical properties in the near-wall region also becomes important for the determination of several non-dimensional numbers (e.g., Ra , Gr , Pr , etc.) involved in the calculation of convective heat transfer.

Alternatively, the calculation of convective heat transfer based on an effective thermal diffusivity approach has been found by the authors to be sensitive to how accurately or not turbulence (hence, sgs quantities like μ_{sgs} and Pr_t), is predicted near the walls. Some tests with such an approach were not particularly successful in predicting grid-insensitive convective heat fluxes in this case.

Acknowledging the above-mentioned limitations, a simplified approach for the determination of the convection coefficient is employed here, which mainly depends on ΔT . The convection heat transfer coefficient, h , is calculated considering simplified empirical correlations for natural convection [30]:

- Laminar flow ($Ra \leq 10^9$):

$$h = 1.42 \left(\frac{\Delta T}{L_c} \right)^{1/4} \quad (13)$$

- Turbulent flow ($Ra > 10^9$):

$$h = 1.31 (\Delta T)^{1/3} \quad (14)$$

where Ra is the Rayleigh number and L_c is the characteristic length (i.e., the height of the walls). It is worth noting that in this case (i.e., natural convection), h is only dependent on L_c in the laminar, and not in the turbulent, flow regime.

The convective heat transfer approach described above is implemented as a boundary condition for the enthalpy equation at the burning surface. When mass

transpiration effects are present (e.g., burning of flammable walls), a reduction in convective heat transfer with increasing mass transfer from the wall can occur (i.e., ‘blowing effect’ [31]). With the current approach, this effect is expected to be implicitly captured through a reduction in the temperature difference involved in the calculation of ΔT . Similar approaches have also been previously employed in [4] which examined the influence of convective heat transfer modelling in CFD simulations of upward flame spread. In accordance with the experiments, convective (considering natural convection) and radiative heat losses are considered at the backside of the MDF / plywood walls while conductive losses are considered for the PMMA walls.

2.5. Pyrolysis modelling

The thermal decomposition of the solid material is modelled using a simplified 1D pyrolysis model [32] which solves conservation equations for mass and energy. A single-step Arrhenius pyrolysis reaction is employed considering three species: virgin, pyrolysate and char (only for charring materials).

- Solid reaction:

$$\text{virgin} = \frac{\rho_c}{\rho_v} \text{char} + \left(1 - \frac{\rho_c}{\rho_v}\right) \text{pyrolysate} \quad (15)$$

- Mass conservation:

$$\frac{\partial}{\partial t}(\rho Y_v) = \dot{\omega}_v''' \quad (16)$$

- Energy conservation:

$$\frac{\partial}{\partial t}(\rho c_p T) = \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \dot{\omega}_v''' \Delta H_{p,v} \quad (17)$$

- Volumetric mass consumption rate:

$$\dot{\omega}_v''' = - \left[\frac{\rho Y_v}{(\rho Y_v)_0} \right]^n (\rho Y_v)_0 A \exp \left(- \frac{E_a}{RT_w} \right) \quad (18)$$

- Mass flow rate of pyrolysis gases per unit area:

$$\dot{m}_{pyr}'' = \left(1 - \frac{\rho_c}{\rho_v}\right) \int_0^W \dot{\omega}_v''' dx \quad (19)$$

where ρ_v and ρ_c are the densities of virgin and char, respectively, Y_v is the mass fraction of virgin material, T is the temperature, $\Delta H_{p,v}$ is the heat of pyrolysis, n is the reaction order, A is the pre-exponential factor, E_a is the activation energy, R is the universal gas constant, W is the wall thickness, the sub-script *pyr* refers to the pyrolysate gas while 0 designates the initial conditions before heating and pyrolysis. Density, ρ , and thermal conductivity, k , are bulk values averaged by the volume fractions of virgin and char, while specific heat, c_p , is averaged based on mass fractions [32]. All properties are assumed to be temperature-independent while the pyrolysate escapes the solid upon formation without any interaction with the solid (i.e., there is no resistance between the pyrolysis gases and the solid phase). The pyrolysis model does not account for water evaporation, moisture mitigation nor in-depth radiation, while cracking and delamination are not modelled, either. All of the above are current limitations of the pyrolysis model that is available in FireFOAM today.

To avoid added complexity, modelling of char oxidation was not considered in the present study, however, its influence will be explored in the future.

The pyrolysate gases are considered to be the same as the fuel in the gas phase (C_3H_8) and the mass loss rate from the walls is calculated as:

$$\dot{m}_w'' = \dot{m}_{pyr}'' \frac{\Delta H_{pyr}}{\Delta H_{c,C_3H_8}} \quad (20)$$

where $\Delta H_{c,C_3H_8} = 46.45 \cdot 10^6$ J/kg is the heat of combustion of C_3H_8 .

The experimental heat flux data considered for validation purposes in this study have been measured using water-cooled Schmidt-Boelter heat flux gauges. Hence, in order to have a direct comparison between the simulations and the experiments, the total gauge heat fluxes in the simulations are determined as:

$$\dot{q}_{gauge}'' = \epsilon_{gauge}(\dot{q}_{inc}'' - \sigma T_{gauge}^4) + \dot{q}_c'' + h(T_w - T_{gauge}) \quad (21)$$

where $\dot{q}_{inc}''$ is the incident radiative heat flux, $\sigma = 5.67 \times 10^{-8}$ W/m²/K⁴ is the Stefan - Boltzmann constant, $\epsilon_{gauge} = 0.97$ while $T_{gauge} = 323$ K [33] for the inert / MDF / plywood scenarios and $T_{gauge} = 291$ K [7] for the PMMA scenarios.

3. Experimental test cases

A brief overview of the different experiments used for validation of the numerical simulations is presented in this section. A stair-step approach is employed by

Table 1: Overview of the 37 cm C₃H₈ pool fire experiments.

Case	20.7 kW	34.4 kW	50.1 kW
$\dot{m}$ (g/s)	0.45	0.74	1.08
$\chi_{r,b}$ (-)	0.23	0.30	0.33
D^* (m)	0.20	0.25	0.29
Q^* (-)	0.22	0.37	0.54
$D^*/\Delta x$ (-)	5 - 20	6 - 25	7 - 29

1 considering gaseous pool fire experiments (only gas phase), anaerobic pyrolysis
 2 experiments (only solid phase) as well as SBI experiments (coupling between the
 3 gas and solid phase) involving both charring and non-charring materials.

4 • Pool fire experiments: The medium-scale C₃H₈ pool fire experiments (i.e.,
 5 20.7, 34.4 and 50.1 kW), involving a 37 cm in diameter circular burner, by
 6 Falkenstein-Smith et al. [34], are considered for validation purposes of the
 7 modelling approach in the gas phase. This series of experiments was cho-
 8 sen because they involve the same fuel and comparable burner size as the
 9 one involved in the SBI scenarios. An overview of some of the most impor-
 10 tant parameters (e.g., heat release rate, radiative fraction and characteristic

11 fire diameter $D^* = \left(\frac{\dot{Q}}{\rho_\infty c_p T_\infty \sqrt{g}} \right)^{2/5}$ involved in the pool fire scenarios is
 12 presented in Table 1.

13 • Pyrolysis experiments: A series of anaerobic pyrolysis experiments are used
 14 for validation purposes of the modelling approach in the solid phase. The
 15 scenarios considered involve polymethyl methacrylate (PMMA) experiments
 16 performed in the NIST Gasification Apparatus [35] and MDF and plywood
 17 experiments performed in the Fire Propagation Apparatus (FPA) [8]. The
 18 combination of both these experiments will enable the validation of the py-
 19 rolysis model in scenarios involving both non-charring and charring mate-
 20 rials. An overview of the different pyrolysis experiments considered (i.e.,
 21 material, sample dimensions and applied heat flux) is presented in Table 2.

22 • SBI experiments: A series of SBI experiments are used for validation pur-
 23 poses of the gas - solid phase coupling modelling approach. The cases
 24 involve inert calcium silicate walls [5], flammable MDF / plywood walls

Table 2: Overview of the pyrolysis experiments.

Sample	Diameter (m)	Thickness (m)	Heat flux (kW/m ²)
MDF	0.0953	0.0184	25, 50, 100
Plywood	0.0953	0.0169	25, 50, 100
PMMA			
Mass-R3	0.07	0.00590	50
Mass-R4	0.07	0.00565	50
Mass-R5	0.07	0.00620	50
Temp-R3	0.07	0.00612	50
Temp-R4	0.07	0.00610	50
Temp-R5	0.07	0.00625	50

[6] and flammable PMMA walls [7]. The choice of these experiments aims to evaluate the predictive capabilities of the employed modelling approach in accurately simulating flame spread in medium-scale scenarios involving both non-charring and charring materials. An overview of the different experiments considered (i.e., material and wall dimensions), along with the main burner characteristics involved in the scenarios, is presented in Table 3.

It should be noted that, while the burner sizes in the pool fire experiments are comparable to the ones used in the SBI experiments, there are differences in their corresponding surface areas. The surface area of the burner will naturally affect the resulting fire plumes (e.g., the flame height and the spatial distribution of radiation). Unfortunately, to the best of the authors' knowledge, no experimental datasets are available for triangular burners with free-burning pool fire flames, so that systematic validation is not possible to date. As such, some uncertainty in the stair-step validation methodology considered in the paper will be present.

4. Numerical setup

A computational domain of 1 m × 2 m × 1 m (L × H × W) is used to model all SBI scenarios (Figure 2). The considered grid sizes include 4 cm (uniform mesh), 2 cm (uniform mesh) and 1 cm (local grid refinement) with the gas phase

Table 3: Overview of the SBI experiments.

Wall	Dimensions
Calcium silicate	1 m x 1.5 m, 0.5 m x 1.5 m
MDF	1 m x 1.5 m, 0.5 m x 1.5 m
Plywood	1 m x 1.5 m, 0.5 m x 1.5 m
PMMA	0.5 m x 1.46 m, 0.5 m x 1.46 m
Burner	
Fuel	C_3H_8
HRR	30 kW
$\chi_{r,b}$	0.33 [34]
Time to peak HRR	30 s
Dimensions	0.25 m x 0.25 m
Shape	Isosceles right triangle

1 cells mapped onto the solid walls. It is worth noting that, for consistency reasons,
 2 the same grid sizes that were employed in the pool fire scenarios are also used
 3 in the simulations of the SBI scenarios. Given the size of the burner involved in
 4 the SBI scenarios (see Table 3), a 4 cm grid size was the coarsest possible that
 5 could be used in the simulations while still maintaining a reasonable resolution of
 6 the fire plume dynamics. Through the thickness of the walls, 30 and 45 cells are
 7 considered for the inert and flammable walls, respectively. The number of cells
 8 inside the solid (i.e., in the direction normal to the wall surface) was chosen to be
 9 equal to $\sqrt{(\tau k / \rho c_p)}$, where τ is a time constant set to 1 s and $k / \rho c_p$ is the thermal
 10 diffusivity [36]. A 30 kW propane triangular burner with a prescribed time ramp
 11 to reach the maximum HRR after 30 s, according to the experiments [3], is the ini-
 12 tial fire source. For the solution of the RTE, 100 solid angles are used for angular
 13 discretization. This choice, which is the default value used by FDS [36], is a com-
 14 promise between computational cost and accuracy. An overview of the setup used
 15 in the numerical simulations is reported in Table 4. Four different wall materials
 16 are considered, including both inert (calcium silicate) and flammable (PMD, ply-
 17 wood, PMMA) surfaces, with their model-effective material properties, reported
 18 in Table 5, derived via inverse modelling and optimization approaches using Fire-
 19 FOAM [8]. For more details regarding the optimization algorithms the reader is
 20 referred to [37]. The optimization was performed by FM Global based on experi-
 21 mental data from PMMA pyrolysis tests in nitrogen environment at external heat

1 fluxes of 25 and 65 kW/m². However, no more specific information regarding the
 2 optimization process is currently available. The C_v values, somewhat higher com-
 3 pared to the ones often found in the literature, most likely account for any limita-
 4 tions of the employed pyrolysis model (e.g., temperature-independent properties
 5 among others) that are often lumped in the determination of the model-effective
 6 parameters. The two sides of the domain, facing the walls, are treated as open'
 7 allowing for air to be entrained. At the top boundary, a zero gradient boundary
 8 condition is assigned to all variables with no reverse flow allowed, to avoid numer-
 9 ical instabilities. The solid surfaces outside the inert / flammable walls, as well as
 10 the floor, are treated as inert surfaces with a no-slip boundary condition applied
 11 for velocity and a small value assigned for sgs kinetic energy, sgs viscosity and
 12 chemical species.

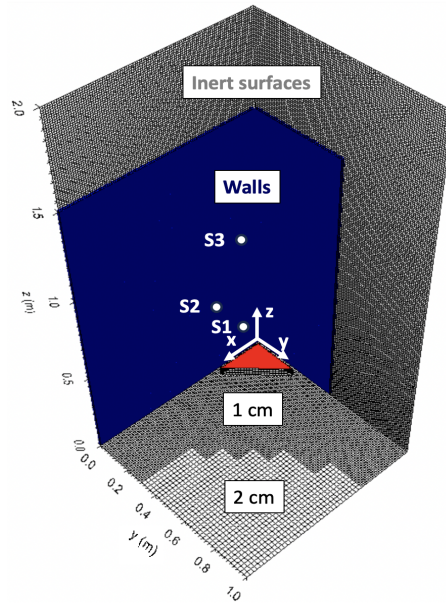


Figure 2: Computational domain and local grid refinement used in the simulations. Blue color: MDF / plywood walls, red color: burner. The location of the gauge heat flux sensors S1 ($x = 0.08$ m, $z = 0.21$ m), S2 ($x = 0.2$ m, $z = 0.35$ m) and S3 ($x = 0.08$ m, $z = 0.8$ m), as well as the orientation axis, are also depicted.

13 The equations are advanced in time using a second order backward scheme.
 14 A second order central difference scheme (linear) is employed for the convective terms while a TVD scheme (limitedLinear 0.25), blending linear and upwind schemes, is used for scalar transport. A central difference scheme (linear) is ap-
 16

Table 4: Overview of the setup used in the simulations.

Domain	1 m x 2 m x 1 m
Grid size (gas)	1 cm (half domain) 2 cm (domain) 4 cm (domain)
Cells (gas/solid)	1.42 mil. / 675 k (1 cm) 250 k / 169 k (2 cm) 31 k / 43 k (4 cm)
Initial conditions	$T_{amb} = 293$ K $P_{amb} = 101325$ Pa
Time step	Variable (max $Co = 0.9$)

1 plied to the diffusive terms while a second order scheme (linearUpwind), which
 2 employs upwind interpolation weights with an explicit correction based on the
 3 local cell gradient, is used for the discretization of the RTE. The choice of the
 4 discretization schemes is based on [38]. The simulations use a variable time step
 5 limited by a maximum Courant number of 0.9. The gas burner scenarios were run
 6 for 35 s (averaged over the last 30 s), the inert wall scenarios were run for 300 s
 7 (averaged over the last 60 s), while the MDF / plywood and PMMA wall scenarios
 8 were run for 300 s and 180 s, respectively.

9 5. Results

10 5.1. Pool fire simulations

11 Figure 3 shows the predicted centerline temperatures and axial velocities,
 12 where z represents the axial distance and D^* is the characteristic fire diameter
 13 (see Table 1). The error bars indicate a 10% uncertainty in the experimental mea-
 14 surements. The numerical predictions demonstrate minimal sensitivity to grid size
 15 and align well with the experimental data, remaining within the uncertainty range
 16 across the considered heat release rates. The simulations effectively capture the
 17 steep temperature gradients and the rapid rise in axial velocities due to buoyancy
 18 observed just above the fire source, regardless of grid resolution. Additionally,
 19 the predicted radiative fractions closely match the experimental values, differing

Table 5: Model-effective material properties^a used in the simulations.

Property	MDF [8]	Plywood [8]	PMMA ^e
W (m)	0.0182	0.017	0.0058
k_v (W/m/K)	0.18	0.12	0.223
ρ_v (kg/m ³)	605	560	998
C_v (J/kg/K)	1580	1330	3147
ϵ_v^b (-)	Varies	Varies	0.85
α_v^b (-)	Varies	Varies	0.9
k_c (W/m/K)	0.21	0.19	-
ρ_c (kg/m ³)	125	82	-
C_c (J/kg/K)	1450	1450	-
ϵ_c (-)	0.86	0.85	-
α_c (-)	0.86	0.85	-
A (s ⁻¹)	$1.47 \cdot 10^4$	$6.30 \cdot 10^4$	$1.59 \cdot 10^{14}$
E_a (J/mol)	$7.19 \cdot 10^4$	$8.01 \cdot 10^4$	$1.84 \cdot 10^5$
n (-)	0.66	2.94	1.0
$\Delta H_{p,v}$ (J/kg)	$1.05 \cdot 10^5$	$-2.15 \cdot 10^5$	$-7.72 \cdot 10^5$
ΔH_{pyr}^c (J/kg)	$8.12 \cdot 10^6$	$6.53 \cdot 10^6$	$29.7 \cdot 10^6$
$\chi_{r,w}^d$ (-)	0.30	0.30	0.36

^aCalcium silicate wall properties taken from [5]: $W = 0.0123$ m, $k_v = 0.17$ W/m/K, $\rho_v = 1005$ kg/m³, $C_v = 920$ J/kg/K, $\epsilon_v = \alpha_v = 0.9$.

^b Temperature-dependent properties which vary based on the polynomial functions reported in [8].

^c Calculated based on the effective heat of combustion of the virgin materials (from cone calorimeter data at 50 kW/m²), ΔH_{eff} (i.e., $12.56 \cdot 10^6$ J/kg for MDF [33], $14.15 \cdot 10^6$ J/kg for plywood [33] and $29.7 \cdot 10^6$ J/kg for PMMA [39]), and the maximum experimentally reported char yield values, Y_c (i.e., 0.18 for MDF [33], 0.29 for plywood [33] while PMMA is assumed to leave no residue). Char is treated as carbon with $\Delta H_{c,c} = 32.8 \cdot 10^6$ J/kg.

^d The values for MDF and plywood correspond to those of pine [19] while the value for PMMA is taken from [40].

^e Optimized from FM Global’s experimental data (FM Global submission of MaCFP 2023 - see https://github.com/MaCFP/macfp-db/tree/master/Fire_Growth/NIST_Parallel_Panel/Computational_Results/2023/FMGlobal).

by only 0.5% to 4%, depending on the grid size. This first validation step, combined with previous validation efforts in numerous buoyancy-driven fire scenarios [1], provides confidence that the employed approach is capable of modelling accurately the gaseous burner which is also a prerequisite for the coupled flame growth simulations presented later.

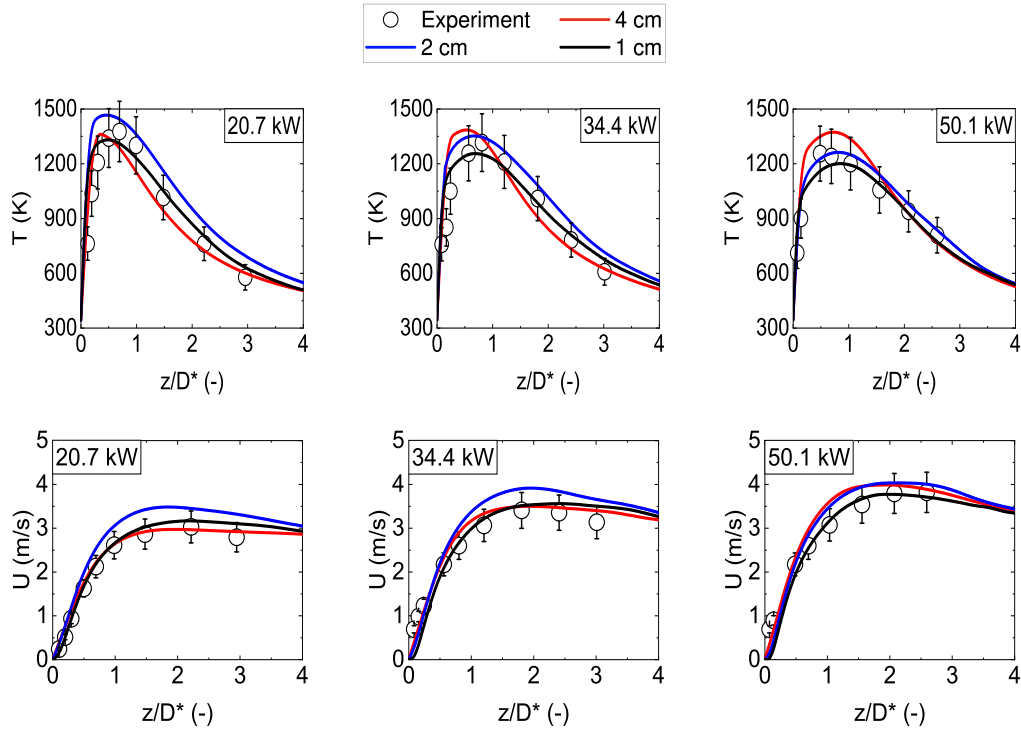


Figure 3: Predicted centerline temperature (top) and axial velocity (bottom) for C_3H_8 pool fires with various heat release rates. Experimental data taken from Falkenstein-Smith et al. [34].

1 5.2. Pyrolysis simulations

2 Figures 4 - 5 present results from pyrolysis simulations of MDF / plywood
3 and PMMA samples, respectively, under different heat fluxes. The objective is to
4 validate the model-effective material properties, previously reported in Table 5,
5 in pyrolysis simulations with well-characterised initial and boundary conditions
6 without the added complexity of having to model the gas phase. It is worth noting
7 that the considered MDF / plywood experiments are the same dataset for which the
8 model-effective material properties were optimised [8], hence, a reasonable match
9 between the simulations and the experiments is to be expected. Nevertheless, the
10 results still aim to validate the pyrolysis model and the FireFOAM code used in
11 the present study which are both somewhat different compared to versions used
12 in the optimization process in [8]. On the other hand, the model-effective material
13 properties for PMMA are applied to experiments which are different compared to
14 the dataset for which they were optimised.

15 Overall, the agreement between the experimental data and the simulation re-

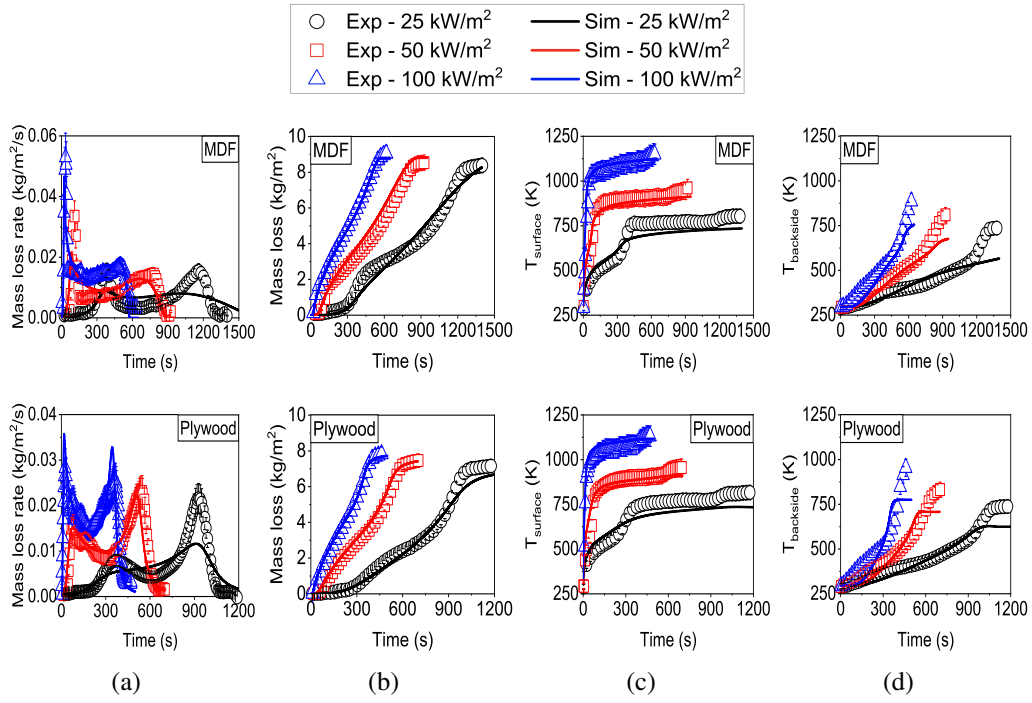


Figure 4: Predicted (a) mass loss rate, (b) mass loss, (c) surface temperature and (d) backside temperature in pyrolysis simulations of MDF (top) and plywood (bottom) samples.

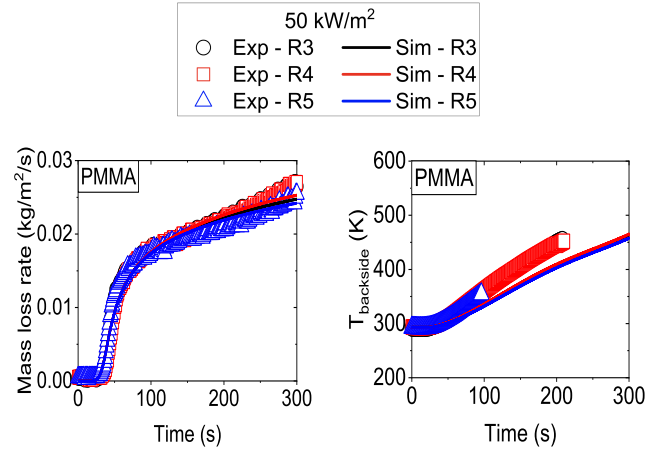


Figure 5: Predicted (a) mass loss rate (cases Mass R3-R5) and (b) backside temperature (cases Temp R3-R5) in pyrolysis simulations of PMMA samples.

sults is reasonable for both MDF and plywood. The simulations effectively capture many of the key experimental trends, including the double peak in the mass loss rate (in the higher heat flux cases) and the rapid rise in the surface temperature. However, some discrepancies are also observed such as e.g., a lower mass loss rate in the second peak and reduced final backside temperatures, particularly at the lower heat flux conditions. Generally, the first peak in the mass loss rate is influenced by the boundary conditions and the applied heat flux on the sample surface, and is typically well captured by the simulations. The second peak, however, occurs after heat has penetrated the char layer formed on the top surface during the early stages of pyrolysis, and can be influenced by the backside boundary conditions. Overall, discrepancies are observed in the predicted backside temperatures towards the end of the tests for both materials. As noted in [8], the exposure of thermocouples and the back-boundary to heater radiation through cracks in the samples may partially explain these discrepancies. Other potential factors contributing to the differences between the simulations and experiments include the assumption of temperature-independent material properties, the pyrolysis model's failure to account for water evaporation or moisture transfer to the backside of the sample, as well as the lack of modelling of cracking or delamination in the samples. Specific to the MDF samples, the consideration of a non-uniform density profile (i.e., parabolic with the exposed surfaces being denser) [41], could also potentially lead to further improvements in the model predictions. In the case of PMMA, the predicted time evolution of the mass loss rate is in reasonable agreement with the experimental data for all the different scenarios considered. The predicted backside temperatures are reasonably well predicted in the first stages of the scenarios (i.e., $t < 100$ s) but start to deviate with increasing time. The discrepancies could be attributed to some of the reasons previously mentioned for the MDF / plywood cases (i.e., related to simplifications on pyrolysis modelling) as well as on the fact that in-depth radiation was not considered in this case. Nevertheless, given the limitations and simplifications used in the pyrolysis modelling in all cases, the model predictions are deemed satisfactory for both charring (MDF / plywood) and non-charring (PMMA) materials. Enhancements in pyrolysis modelling in the future are expected to lead to further improvements in the current model predictions.

5.3. Inert wall simulations

Figure 6 shows the average heat release rate per unit volume (isocontour of 200 kW/m³) for different grid sizes in the simulations. Overall, for the coarser 4 cm grid, the flames are slightly shorter and larger in volume near the base of the burner

1 due to lack of turbulence. Refining the grid to 2 cm and 1 cm results in a more
2 turbulent flame, which increases in length and becomes narrower near the base of
3 the burner due to stronger air entrainment. It is believed that further improvements
4 in turbulence modelling (i.e., focusing on the reverse energy cascade that typically
5 occurs near the base of the burner) and combustion modelling (i.e., refining the
6 mixing time scale on coarse grids) could further enhance the flame structure when
7 using coarser grids.

8 A comparison between the predicted total heat fluxes in the inert wall scenar-
9 ios is presented in Figure 7. A 2 sec running averaging has been applied to the
10 data in order to smoothen their fluctuations and to make the comparison with the
11 experimental data easier. These scenarios aim to evaluate whether the considered
12 modelling approach can accurately predict the heat feedback (due to convection
13 and radiation) to the wall surface, which is important in terms of predicting the
14 fuel gasification in the flame growth scenarios presented later. It is worth noting
15 that the reported experimental flame height for these inert wall scenarios was ap-
16 proximately 0.87 m [5]. Based on this, as well as on reported photos from the
17 experiments (see [5]), sensor S1 is located within the flame region (i.e., it is con-
18 stantly engulfed by flames from the burner), sensor S2 is located on the side of
19 the burner (i.e., where flames are only occasionally present) while sensor S3 is
20 located in the intermittent region (i.e., where flames are very often present).

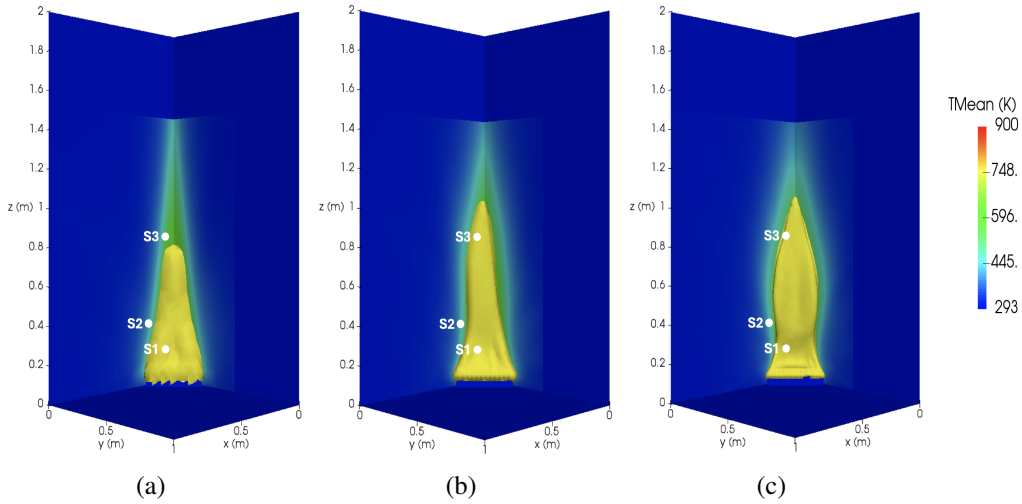


Figure 6: Average heat release rate per unit volume (isocontour of 200 kW/m³) and inert wall temperatures for a grid size of (a) 4 cm, (b) 2 cm and (c) 1 cm.

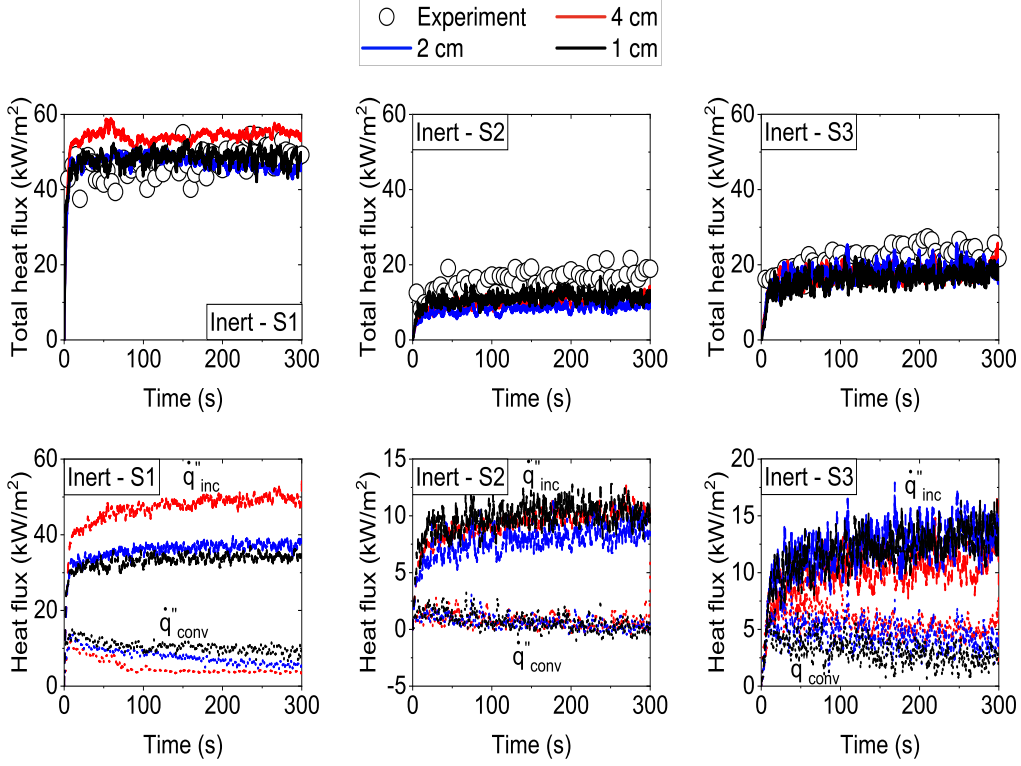


Figure 7: Total heat fluxes at sensors S1, S2 and S3, as a function of grid size, in the inert wall simulations.

1 The predicted radiative fractions from the simulations are approximately 33.2%,
 2 32.9%, and 33.8% for the 4 cm, 2 cm, and 1 cm grid sizes, respectively, all of
 3 which are very close to the target value of 33%. This indicates that the employed
 4 modelling approach ensures the correct amount of radiation is released, regardless
 5 of the grid size, which is essential for accurately predicting heat feedback on the
 6 wall surface. However, as will be discussed later, this factor alone is not sufficient
 7 for accurately predicting heat feedback to the fuel surface and subsequent upward
 8 flame spread. The flame heights in the simulations are estimated using the cri-
 9 terion of integrated heat release rate per unit volume (97% in this case which is
 10 an average value within the 95-99% range often used in the literature [42]), with
 11 predicted values of approximately 0.74 m, 0.87 m, and 0.89 m for the 4 cm, 2
 12 cm, and 1 cm grid sizes, respectively. Thus, the 2 cm and 1 cm grid simulations
 13 closely match the experimentally reported flame height of 0.87 m [5], while the 4
 14 cm grid size scenario remains relatively close.

1 The differences in flame structure as a function of grid size (i.e., flames with
2 larger volume and more concentrated near the corner with the 4 cm grid) are sig-
3 nificant and locally affect the spatial distribution of the heat feedback to the fuel
4 surface. This is further illustrated by examining the total and individual heat fluxes
5 at sensors S1, S2, and S3, as shown in Figure 7. In particular, higher incident heat
6 fluxes, ranging from 10 to 15 kW/m², are observed at sensor S1 for the coarsest
7 grid size (4 cm) compared to the finer grid sizes (2 cm and 1 cm). This could be
8 partially attributed to the locally higher maximum flame temperatures predicted in
9 the coarser grid simulations, as seen in the burner scenarios (which are relatively
10 close to the burner surface, as shown in Figure 3).

11 On the other hand, the predicted convective heat fluxes at sensor S1 are not
12 highly sensitive to the grid size, exhibiting an increasing trend as the grid is re-
13 fined. The 4 cm grid, despite being relatively coarse, predicts convective heat
14 fluxes initially around 10 kW/m², which decrease to approximately 5 kW/m² to-
15 wards the end of the simulation as the walls heat up and ΔT decreases. For
16 the finer grid sizes (2 cm and 1 cm), which resolve the boundary layer near the
17 wall more accurately, the predicted convective heat fluxes are higher, i.e., initially
18 around 15 kW/m² and decreasing to about 5 kW/m² by the end of the simulation,
19 for the same reason. This also leads to a slight reduction in the total heat fluxes
20 observed at this location over time.

21 At sensor S2, no significant differences in the predicted heat fluxes as a func-
22 tion of grid size are observed, as flames are only intermittently present in this area,
23 with all simulations showing similar trends. Overall, the total heat fluxes are more
24 accurately predicted with the finest grid size (1 cm), likely due to the increased
25 flame turbulence. However, the predicted total heat fluxes remain under-predicted
26 by all grid sizes by about 5 kW/m² compared to experimental data. Minor dif-
27 ferences are also observed at sensor S3, with total heat fluxes being close to the
28 experimental measurements for all grid sizes (which is not surprising, given the
29 comparable flame heights predicted in all cases).

30 Nevertheless, simply examining total heat flux predictions at three distinct lo-
31 cations on the wall surface is insufficient for evaluating whether the simulations
32 accurately predict the heat feedback to the fuel surface. Ideally, a grid of heat flux
33 measurements across the entire wall surface would be necessary to make conclu-
34 sive statements about the accuracy of the CFD predictions. Therefore, considering
35 additional inert panel scenarios with more available heat flux measurements will
36 be important for future studies.

37 Overall, for the reasons discussed above, grid-insensitive predictions for to-
38 tal heat fluxes are achieved only with grid sizes of 2 cm and 1 cm. In contrast,

1 the 4 cm grid size simulations tend to over-predict the experimental data. It is
 2 important to note that total heat fluxes are often under-predicted in coarse grid
 3 simulations, either because radiative fractions (and thus radiative heat fluxes) are
 4 under-predicted, or because convective heat fluxes are not well resolved (since
 5 they rely on accurately capturing the temperature gradient near the surface). How-
 6 ever, this is not the case in the present simulations, regardless of the grid size (1 - 4
 7 cm). For all considered grid sizes, the correct amount of radiative heat is released
 8 globally (Eqs (9)-(10)), and the convective fluxes do not exhibit significant grid
 9 dependency, even though the grid sizes are not particularly fine for resolving the
 10 boundary layer near the wall surface.

Table 6: Steady-state heat feedback (kW), as a function of grid size, in the inert wall simulations.

Case	Q_{inc} (kW)	Q_{conv} (kW)	Q_{total} (kW)
4 cm	10.75	1.62	12.37
2 cm	10.78	1.78	12.57
1 cm	11.15	1.64	12.80

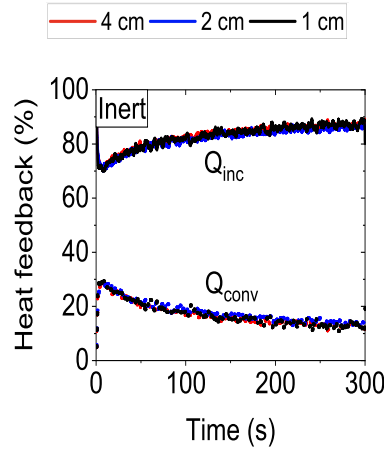


Figure 8: Transient heat feedback (%), as a function of grid size, on the inert wall simulations.

11 The numerical predictions of the heat feedback (kW) during steady-state con-
 12 ditions (calculated over the last 60 s in the simulations), reported in Table 6, as
 13 well as the transient heat feedback (%) on the wall surface, shown in Figure 8, fur-

ther verify some of the above-mentioned findings and help with the further analysis of the results. The total heat feedback received by the walls slightly increases with decreasing grid size (i.e., due to better resolution of the flame structure and temperature). When focusing on the finer grid size (2 cm and 1 cm) scenarios, convection accounted for approximately up to 30% (while radiation accounted for 70%) of the total heat feedback in the early stages of the scenarios where pre-heating of the inert walls occurs. This amount reduced to approximately 15% (and 85% for radiation) during steady-state conditions since the walls have had sufficient time to heat up and the amount of convective heat transfer reduces.

The results presented above indicate that coarser grid simulations (4 cm) can capture some of the main flow characteristics and flame dynamics associated with these scenarios and that, when compared to the finer grid simulations (2 cm and 1 cm), they can capture, both qualitatively and quantitatively, reasonably well the heat feedback to the fuel surface, which is important for accurate predictions of flame spread. With the current approach, the use of coarse grids (i.e., often used for engineering type of simulations), however, might lead to higher total heat fluxes to the fuel surface at locations close to the original fire source. Hence, attention is required when heat transfer is of interest, e.g., in scenarios involving flame spread and / or liquid evaporation scenarios. It is worth noting that measurements from well-characterised experiments at different scale, reporting both convective and radiative heat fluxes (not just total heat fluxes), are ideally needed for further future model validation.

5.4. *Flammable wall simulations*

Figure 9 presents a qualitative transient evolution of the instantaneous heat release rate per unit volume (isocontour of 200 kW/m³) and wall temperatures for the plywood walls for two different grid sizes (4 cm and 1 cm). In this case, the 4 cm grid size is too coarse and results in a flame that is not turbulent enough and lacks any detailed structure. On the other hand, the use of a finer grid size (1 cm) leads to a more well-resolved flame structure and clear vortices are observed. In the case of the 1 cm grid size, weakly-turbulent flame structures are evident near the base of the walls and clearly illustrate the transition of the flow from quasi-laminar, just above the burner, to turbulent, further downstream towards the top of the walls. Similar observations were made in the scenarios involving the MDF and PMMA walls as well (not presented here to avoid repetition).

The predicted total heat release rates, which include contributions from both the burner and the flammable walls, are shown in Figure 10. The total heat release rate is calculated by integrating the local heat release rate per unit volume

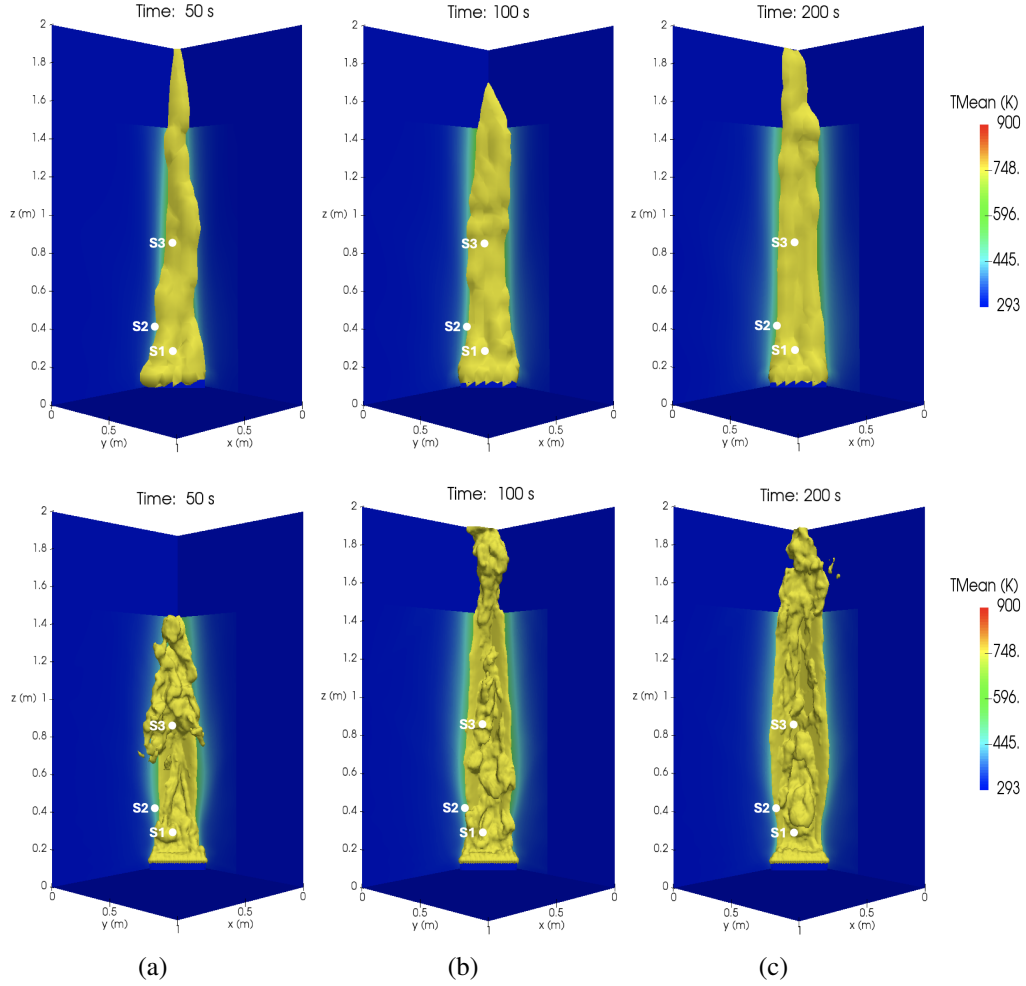


Figure 9: Instantaneous heat release rate per unit volume (isocontour of 200 kW/m^3) and plywood wall temperatures for a grid size of 4 cm (top) and 1 cm (bottom) at time $t =$ (a) 50 s, (b) 100 s and (c) 200 s.

1 in every cell across the entire computational domain. Despite the use of different
 2 grid sizes (ranging from 1 - 4 cm), the heat release rate predictions are notably
 3 grid-insensitive and generally compare well against the experimental data in most
 4 cases. It should be noted that the choice of coarsest grid size in the simulations
 5 was somewhat restricted by the size of the burner (i.e., isosceles right triangle with
 6 a side length of 0.25 m) involved in the scenarios. The use of a 4 cm grid size,
 7 resulting in approximately 6 cells across the side length of the burner, was the

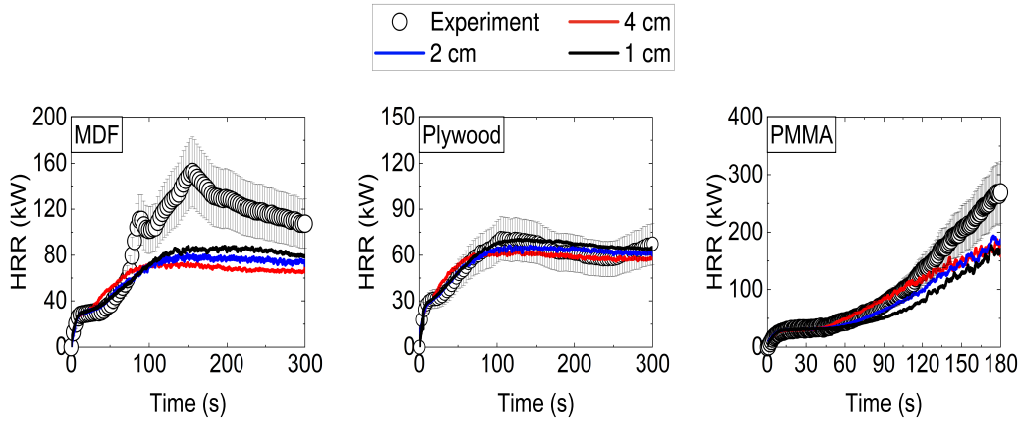


Figure 10: Total heat release rate (burner and walls), as a function of grid size, in the flammable wall simulations. The experimental data for the MDF / plywood scenarios represent an average of three tests.

1 minimum allowed in order to still have a reasonable resolution of the fire plume
2 dynamics.

3 As expected, higher maximum heat release rates are obtained with decreasing
4 grid size due to better resolution of the flame structure and temperature which
5 increases the overall heat feedback to the wall surface (previously also observed
6 in Table 6 for the inert wall scenarios). Nevertheless, the effect of having (slightly)
7 higher HRR, due to over-predicting the total heat fluxes near the burner on coarse
8 grids as previously discussed in the inert wall cases, is evident in the early stages
9 of these flame spread scenarios. The gradual increase in the HRR until it reaches
10 a plateau at approximately 30 kW, which is observed in all scenarios, is due to
11 the C_3H_8 burner alone. During this period there is pre-heating (due to convection
12 and radiation) of the flammable walls which eventually leads to pyrolyzing gases
13 being released, the walls to be ignited and the flames to eventually spread both
14 axially and laterally at different rates. The best predictions, when compared to
15 the experimental data, are obtained with the finest grid size (1 cm) used in the
16 simulations for all of the scenarios considered. However, the 2 cm and 4 cm grid
17 size predictions are also rather satisfactory. The maximum absolute deviations
18 in the predicted HRR between the finest (1 cm) and coarsest grid size (4 cm), at
19 any time during the simulations, were approximately 15%, 29% and 39% for the
20 plywood, MDF and PMMA scenarios, respectively. On the other hand, the relative
21 differences in the predicted HRR between the simulations and the experiments
22 occurred at different time intervals and remained less than 47% (see Table 7).

Table 7: Average relative differences (%) at different time intervals in the predicted HRR between simulations and experiments.

	0 - 50 s	50 - 100 s	100 - 200 s	200 - 300 s
MDF				
4 cm	20	-5	-47	-40
2 cm	11	-14	-41	-35
1 cm	11	-16	-37	-29
Plywood				
4 cm	11	1	-9	-4
2 cm	3	-2	-4	2
1 cm	3	2	4	8
PMMA				
4 cm	2	11	-20	-
2 cm	2	-13	-26	-
1 cm	1	-26	-37	-

Part of the discrepancies observed in the MDF scenarios (i.e., under-prediction of the peak HRR) can be attributed to the uniform density profile that was used in the simulations. Even though reasonable predictions were obtained in the small-scale pyrolysis experiments (previously presented in Figure 4 and also reported in [8]), the inclusion of a non-uniform density profile for MDF walls can potentially be important when simulating medium-scale SBI experiments [3]. More specifically, the parabolic in nature profile of the MDF samples (with higher density on the surfaces and lower density in the middle) can eventually lead to a higher HRR in the early stages of flame spread when simulating such scenarios. This aspect has been previously presented in [3] which reported higher heat release rates in the early stages of flame spread during MDF when a variable, as opposed to a constant, density profile was employed in the simulations. This effect will be considered when modelling these scenarios in the future.

On the other hand, a perfect agreement is observed in the case of plywood with the model predictions being rather grid-independent and following closely the experimental measurements. A reasonable agreement is also observed in the case of the PMMA walls as well with the simulations following the experimental trends and predicting a gradual increase in the HRR as a function of time. Nevertheless,

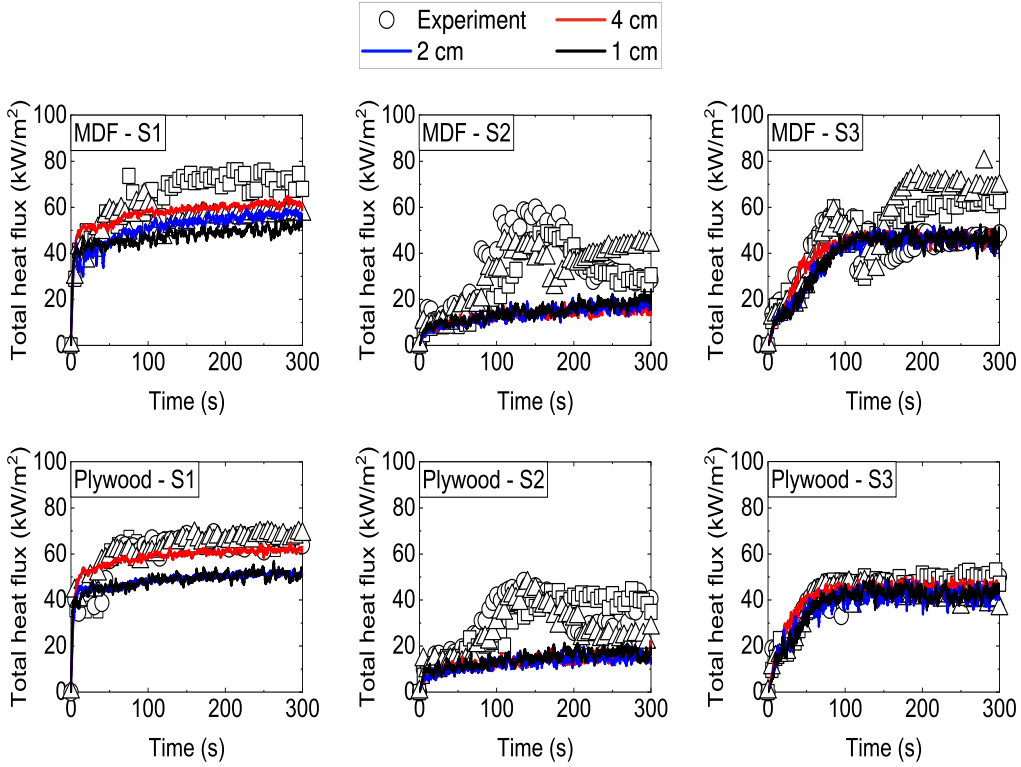


Figure 11: Total heat fluxes at sensors S1, S2 and S3, as a function of grid size, in the flammable wall simulations with MDF (top) and plywood (bottom). The various symbols (i.e., squares, circles, triangles) represent different experiments.

1 the peak value of the HRR around $t = 180$ s is somewhat missed by the simu-
2 lations. This could be linked to discrepancies in the (convective / radiative) heat
3 transfer modelling in the early stages of the scenarios (i.e., being important with
4 respect to pre-heating of the walls). In addition, in the flammable walls involv-
5 ing charring materials (MDF, plywood) some uncertainties in the determination
6 of the heat of pyrolysate also exists, which can be important with respect to the
7 heat feedback and burning of the walls in the gas phase. This aspect will also be
8 further investigated in the future.

9 To aid in the analysis of the results, the total heat fluxes on the MDF / plywood
10 walls (measured at sensors S1, S2, and S3) and on the PMMA walls (at selected
11 locations relatively close to those of sensors S1 - S3) are presented in Figures 11
12 and 12, respectively. A 2-second running average has been applied to smoothen
13 the data fluctuations and make the comparison to experimental data easier. In the

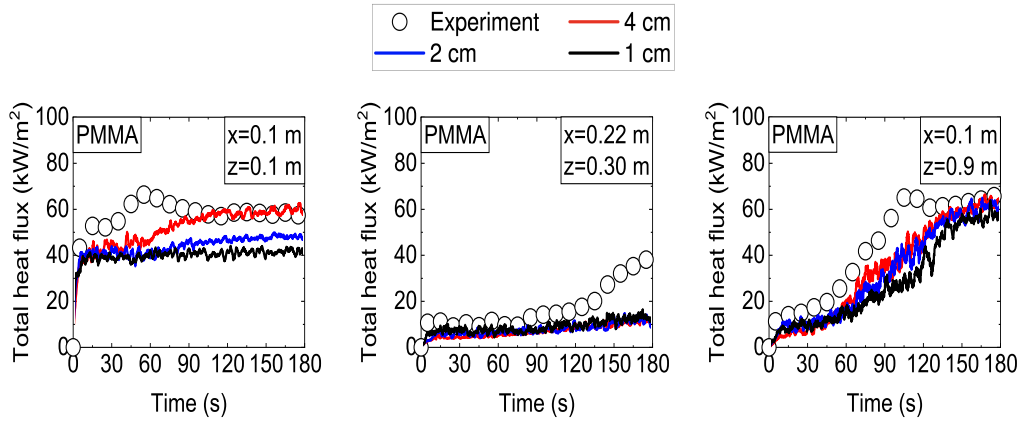


Figure 12: Total heat fluxes, as a function of grid size, at selected locations (somewhat comparable to those of sensors S1, S2 and S3) in the flammable wall simulations with PMMA.

1 MDF / plywood wall scenarios, there is notable scatter in the experimental data
2 for the total heat fluxes, particularly in the MDF case. Specifically, the total heat
3 fluxes for MDF at sensor S1 range from 55 to 75 kW/m², and at sensor S3 from 45
4 to 70 kW/m². In contrast, the experimental measurements for the plywood walls
5 exhibit less scatter, particularly at sensors S1 and S3. Due to this variation, a quan-
6 titative comparison will not be attempted, and instead, a qualitative comparison of
7 total heat flux predictions will be made.

8 For the MDF wall scenarios, the simulation results align well with the major
9 trends observed in the experimental measurements at sensors S1 and S3. At sensor
10 S1, a rapid increase in total heat flux is observed, as flames are consistently present
11 in this location from the beginning of the simulations. The predicted maximum
12 total heat flux is around 60 kW/m², which closely matches the values reported in
13 [3] and aligns with one of the two available experimental data sets at this location
14 (represented by triangle symbols). It is also worth noting that experimental mea-
15 surements for the PMMA walls (evaluated at a similar height z and interpolated
16 between $x = 0.1 - 0.2$ m) suggest total heat flux values of approximately 55 to
17 60 kW/m² during the early stages of the SBI tests (before the PMMA burning
18 begins).

19 A gradual increase in total heat flux is predicted at sensor S3, initially driven
20 by convection (pre-heating the wall) until the flames grow and radiation becomes
21 the dominant heat transfer mechanism. The maximum values predicted by the
22 simulations are approximately 45 kW/m², which are somewhat lower than the
23 range of 50 to 55 kW/m² reported in [3] for uniform density materials, and also on

1 the lower end of the available experimental measurements. This lower prediction
2 is consistent with the fact that fire growth is not accurately captured in the MDF
3 wall simulations, leading to less intense burning and lower heat flux values.

4 At sensor S2, a steady, gradual increase in total heat flux is also predicted, but
5 the distinct peak observed in the experimental data (due to lateral flame spread and
6 the flame front passing through this location) is not captured in the simulations.
7 Additionally, the decrease in total heat fluxes observed when refining the grid size
8 from the coarser (4 cm) to the finer (2 cm and 1 cm) grids is consistent with the
9 findings discussed earlier for the inert wall simulations.

10 More specifically, the decrease in the total heat fluxes is due to a decrease in
11 the incident heat fluxes at this location (previously also seen in Figure 7 and a
12 consequence of the differences in the flame structure). In this case, there is no
13 compensating effect due to convective heat transfer since the walls, at this loca-
14 tion, start burning very fast resulting in comparable ΔT (i.e., hence, also convec-
15 tive heat fluxes) regardless of the grid size. Very similar observations can be made
16 about the numerical predictions in the case of the plywood walls.

17 A notable difference in these scenarios, compared to the MDF wall simula-
18 tions, is that the total heat fluxes from the simulations with the finer grid sizes (2
19 cm and 1 cm) seem to be under-predicted at sensor S1. The experimental measure-
20 ments in this case lie between 60 - 70 kW/m². These values are actually higher
21 than the measurements for the MDF walls, although the MDF burned more inten-
22 sively, as observed from the reported total HRR (i.e., considering that the current
23 and previously reported [3] total heat flux predictions are around 60 kW/m² and
24 also in line with one set of experimental measurements). Given these aspects, it
25 should not be excluded that there might be some discrepancies in the experimental
26 measurements for the total heat flux of the plywood walls at sensor S1.

27 Very similar findings are drawn by the numerical predictions for the total heat
28 fluxes in the case of the PMMA walls which are not listed here to avoid repeti-
29 tion. A notable observation is again the tendency of the simulations to somewhat
30 under-predict the total heat fluxes at low elevations (i.e., at height $z = 0.1$ m - an
31 axial location which is very close to the floor and even lower compared to sensor
32 S1 in the MDF / plywood wall scenarios). Further investigation to some of the
33 modelling choices in the simulations (e.g., related to near wall modelling, convec-
34 tive heat transfer in the laminar regime as well as the choice in the number of solid
35 angles) will be made in the future in order to investigate whether improvements
36 can be made in the total heat fluxes predictions at these low elevations.

37 To further quantify the accuracy of the numerical predictions, the relative dif-
38 ferences (%) in the predicted total heat fluxes between the simulations and the

1 experiments, for different time intervals, are presented in Table 8.

Table 8: Average relative differences (%) for MDF / plywood / PMMA at different time intervals in the predicted total heat fluxes between simulations and experiments.

	0 - 50 s	50 - 100 s	100 - 200 s	200 - 300 s
S1				
4 cm	194 / 79 / -19	49 / 34 / -17	25 / 21 / 3	57 / 28 / -
2 cm	138 / 51 / -23	29 / 10 / -32	13 / 0 / -18	45 / 6 / -
1 cm	154 / 48 / -29	23 / 8 / -35	2 / -1 / -29	30 / 7 / -
S2				
4 cm	-67 / -65 / -60	-71 / -70 / -49	-73 / -73 / -60	-72 / -71 / -
2 cm	-68 / -70 / -43	-75 / -74 / -40	-73 / -71 / -59	-70 / -71 / -
1 cm	-63 / -60 / -21	-70 / -71 / -23	-72 / -70 / -53	-66 / -67 / -
S3				
4 cm	-15 / 3 / -59	0 / 4 / -38	-3 / -6 / -11	-17 / 6 / -
2 cm	-34 / -17 / -39	-13 / -9 / -44	-5 / -14 / -13	-17 / -3 / -
1 cm	-35 / -20 / -46	-18 / -4 / -52	-5 / -11 / -24	-18 / -2 / -

2 The burning area of the (plywood, MDF, PMMA) panels is concentrated where
3 flames also exist in the gas phase and mainly occurs in the vertical direction. The
4 consideration of only 1D heat transfer through the walls does not allow for any
5 significant lateral flame spread (i.e., outside the areas where flames are present)
6 to occur during the simulations. This is believed to be the main reason for any
7 discrepancies observed in the CFD prediction at sensor S2. The under-predictions
8 observed in the burning of the MDF and the PMMA walls during the later stages
9 (i.e., HRR in Figure 10) as well as in the total heat fluxes at sensors S1 and S3
10 (i.e., see Figures 11 - 12) are essentially linked. It can be observed that during
11 the very early stages of the tests (i.e., when pre-heating occurs without yet any
12 significant burning) the total heat fluxes are reasonably well predicted.

13 However, given the positive feedback loop present in such scenarios, if the
14 burning is under-predicted, so are the heat fluxes, and the other way around. Es-
15 pecially in these two scenarios, it is believed that it is modelling deficiencies in the
16 solid phase (i.e., related to the non-uniform density for MDF, potentially with in-
17 depth radiation for PMMA, among others) that lead to the observed discrepancies.
18 The observed differences in the burning between the plywood and MDF walls can

also be linked to the higher char fraction the former have (i.e., making them more difficult to burn) and the non-uniform density effect in the latter (i.e., which was not considered in the modelling and could lead to more intense burning in the earlier stages of burning) as well differences in their optimised thermophysical properties (i.e., particularly the heat of pyrolysis).

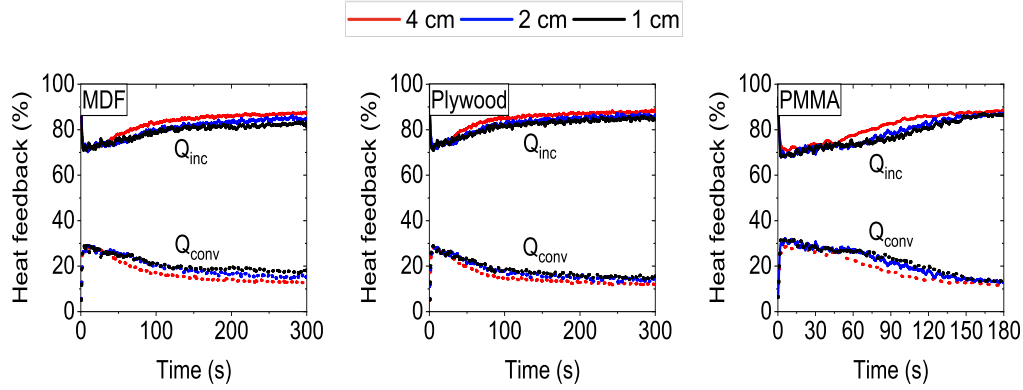


Figure 13: Transient heat feedback (%), as a function of grid size, in the flammable wall simulations.

The numerical predictions regarding the transient heat feedback (%) on the wall surface for the flammable wall simulations is reported in Figure 13. Overall, the heat feedback due to convection is on the order of 30% (while radiation accounts for 70%) in the early stages of flame spread, when pre-heating of the walls occurs, which subsequently reduces to approximately 10 - 20% (while radiation accounts for 80 - 90%), when the flames grow in size and radiation becomes dominant. These values are somewhat quantitatively comparable to what has been reported in the literature in the past during steady burning of PMMA walls [43] (i.e., convection: 20 - 25% and radiation: 75 - 80% above 0.76 m in a 1.57 m high wall). These values suggest that convective heat transfer is not negligible in the early stages of flame spread scenarios and modelling of convection can be important with respect to accurately capturing the pre-heating of the walls and the subsequent flame spread. Further research on convective heat transfer modelling in scenarios involving flammable walls is required in order to determine whether alternative approaches (e.g., based on mass transfer and the use of stagnant film theory [4, 29]) could also be considered.

6. Conclusions

The paper presented large eddy simulations of upward flame spread with Fire-FOAM in Single Burning Item (SBI) scenarios. A fully dynamic approach with respect to turbulence, combustion and radiation was employed, which minimized the use of model constants and aimed at predicting the radiative fractions. Convective heat transfer was modelled based on Newton's law and calculated the convective coefficients considering simplified correlations for natural convection. Solid decomposition was modelled using a 1D model involving a single-step Arrhenius pyrolysis reaction. The approach was extensively validated by simulating the gas (i.e., pool fires) and solid (i.e., anaerobic pyrolysis) phases separately as well as coupled simulations of medium-scale SBI scenarios considering both inert (calcium silicate) and flammable walls involving charring (MDF, plywood) and non-charring (PMMA) materials. The study explicitly focused on flame spread simulations involving grids that are sufficiently small to capture the fire plume dynamics but that are not fine enough to resolve the boundary layer.

The main conclusions drawn from this numerical study are summarized below:

- Given the complexity of the problem, the employed dynamic modelling approach was able to get reasonable and rather grid-independent predictions for the heat release rate (i.e., maximum deviations in the predicted HRR, between the coarsest and finest grid size, on the order of 15-39%) and total heat fluxes on all grid sizes (i.e., in the range of 1 - 4 cm considered in the paper) for a range of scenarios and materials. On the other hand, the relative differences (%) in the predicted HRR between simulations and experiments varied up to 47%, depending on the time interval examined.

Overall, the performance of the approach was very satisfactory and illustrated the potential of CFD to accurately model upward flame spread scenarios even if the boundary layer is not (fully) resolved in the simulations.

- The convective component of the total heat flux at the burning surface was found to be important in the early stages of flame spread (i.e., during pre-heating of the walls) and accounted for approximately up to 30% of the total heat feedback to the wall surface. This value reduced to approximately 10 - 20% in the later stages of flame spread when the fire size increased and radiative heat transfer became dominant. This finding suggests that accurate modelling of convective heat transfer at the burning surface in the early stages can be as important as that of radiation in scenarios involving upward flame spread.

- 1 • Accurate prediction of convective (i.e., grid-insensitive convective heat fluxes)
2 and radiative (i.e., ensuring that the correct radiative fraction is recovered)
3 heat transfer, as a function of grid size, was found to be important for ac-
4 curate (both qualitatively and quantitatively) flame spread predictions using
5 CFD. The use of coarse grids (4 cm in this case) led to an over-prediction
6 of the incident heat fluxes, while the convective heat fluxes were fairly grid-
7 insensitive. More specifically, the coarseness of the grid increased the flame
8 volume near the base of the walls, and combined with the correct emission
9 of heat due to radiation, increased the incident heat fluxes in locations in
10 the vicinity of the burner (which led to over-predicted total heat fluxes as
11 well). This finding suggests that accurate predictions of flame spread can
12 be very challenging for grid sizes typically used for engineering-type of
13 simulations.
- 14 • The considered methodology enabled to separately validate the gas and
15 solid phases. However, some discrepancies were still present in the cou-
16 pled simulations. Possible modelling deficiencies in the gas (e.g., use of
17 fast chemistry) and solid (e.g., no modelling of water evaporation, crack-
18 ing or delamination, temperature-independent properties, constant density
19 variation inside the solid) phases could be responsible for amplifying the
20 errors in the model predictions in the coupled simulations. Even though the
21 current approach performed very satisfactorily in this case, further research
22 and model validation in smaller and larger scale experiments is needed in
23 the future.

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28 **References**

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